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1. Problem Statement

Analyze the Amazon Fine Food Reviews dataset using Hadoop MapReduce and Apache Pig in order to produce useful summaries, rankings, rating-level insights and visualizations. The project uses Hadoop Distributed File System (HDFS) for storage, Java MapReduce for three required analyses, and Apache Pig for five higher-level analytical tasks. The final report follows the same laboratory-report structure as the previous YouTube Hadoop report, but all analysis, screenshots and results in this report are from the Amazon review project.

2. Dataset Description

Dataset source: <https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews>

Field	Description
Id	Unique review identifier.
ProductId	Amazon product identifier.
UserId	Unique reviewer identifier.
ProfileName	Reviewer profile name.
HelpfulnessNumerator	Number of users who found the review helpful.
HelpfulnessDenominator	Total helpfulness votes.
Score	Review rating from 1 to 5.
Time	Unix timestamp of the review.
Summary	Short review summary.
Text	Full review text.

The original CSV contains 568,454 review records. Because review summaries and full text may contain commas, quotation marks and line breaks, the dataset was converted into a tab-separated file before Hadoop processing. The preprocessing stage produced 568,452 valid records and removed 2 invalid records. The cleaned amazon_reviews.tsv file size is 299,117,356 bytes.

3. Summary of Data Analysis

The following three MapReduce analyses were completed:

- Best Reviewer Based on Number of Reviews.
- Total Amazon Reviews by Rating Score.
- Total Helpful Votes Based on Product ID.

The following five Apache Pig analyses were completed:

- Top 5 Products by Number of Reviews.
- Top 10 Most Helpful Reviews.
- Top 10 Helpful Reviews by Rating Score.
- Top 10 Products by Total Helpful Votes.
- Top 10 Products by Helpful Votes for Each Rating Score.

4. Environment and Dataset Preparation

Hadoop Environment

Hadoop services were started successfully on Windows. The JPS output confirms NameNode, DataNode, ResourceManager and NodeManager. JobHistoryServer was also started later for Pig execution.

```
C:\Windows\System32>cd C:\hadoop\sbin  
  
C:\hadoop\sbin>.\start-dfs.cmd  
  
C:\hadoop\sbin>.\start-yarn.cmd  
starting yarn daemons  
  
C:\hadoop\sbin>jps  
20608 NodeManager  
25776 NameNode  
8564 ResourceManager  
8692 Jps  
19304 DataNode
```

Hadoop services confirmed through JPS.

Project Directory Structure

The project was organized into dataset, preprocessing, MapReduce source, compiled classes/JARs, Pig scripts, result, screenshot, visualization and report folders.

```

C:\Users\shari\Amazon-Fine-Food-Hadoop>tree /F
Folder PATH listing for volume Windows
Volume serial number is 000000DC 6207:0771
C:
├── dataset
│   ├── Reviews.csv
│   └── Reviews.csv.zip
├── jar
├── mapreduce
│   ├── best-reviewer
│   ├── product-helpfulness
│   └── rating-count
├── pig
├── preprocessing
├── report
├── results
│   ├── mapreduce
│   └── pig
├── screenshots
│   ├── mapreduce
│   ├── pig
│   └── setup
└── visualization

C:\Users\shari\Amazon-Fine-Food-Hadoop>dir dataset
Volume in drive C is Windows
Volume Serial Number is 6207-0771

Directory of C:\Users\shari\Amazon-Fine-Food-Hadoop\dataset

09/12/2026  12:43 AM    <DIR>          .
09/12/2026  12:42 AM    <DIR>          ..
09/12/2026  12:43 AM           300,904,694 Reviews.csv
09/12/2026  12:24 AM           120,062,355 Reviews.csv.zip
               2 File(s)      420,967,049 bytes
               2 Dir(s)  17,853,501,440 bytes free

```

Project directory structure created for the Amazon Hadoop project.

Cleaned Dataset Sample

The cleaned TSV file keeps the ten expected fields in a Hadoop-friendly tab-separated format. The following screenshot shows the first five cleaned records.

```
C:\Users\shari\Amazon-Fine-Food-Hadoop>powershell -Command "Get-Content dataset\sample_amazon_reviews.tsv -TotalCount 5"
1      B001E4KFG0      A3SGXH7AUHU8GW  delmartian      1      1      5      1
303862400      Good Quality Dog Food  I have bought several of the Vitality
canned dog food products and have found them all to be of good quality. The pr
oduct looks more like a stew than a processed meat and it smells better. My La
brador is finicky and she appreciates this product better than most.
2      B00813GRG4      A1D87F6ZCVE5NK  dll pa 0      0      1      134697
6000      Not as Advertised      "Product arrived labeled as Jumbo Salted Peanu
ts...the peanuts were actually small sized unsalted. Not sure if this was an e
rror or if the vendor intended to represent the product as ""Jumbo""."
3      B000LQOCH0      ABXLMWJIXXAIN  "Natalia Corres ""Natalia Corres"" 1
1      4      1219017600      """"Delight"" says it all"      "This is a con
fection that has been around a few centuries. It is a light, pillowy citrus ge
latin with nuts - in this case Filberts. And it is cut into tiny squares and t
hen liberally coated with powdered sugar. And it is a tiny mouthful of heaven.
Not too chewy, and very flavorful. I highly recommend this yummy treat. If yo
u are familiar with the story of C.S. Lewis' ""The Lion, The Witch, and The Wa
rdrobe"" - this is the treat that seduces Edmund into selling out his Brother
and Sisters to the Witch."
4      B000UA0QIQ      A395BORC6FGVXV  Karl      3      3      2      130792
3200      Cough Medicine  If you are looking for the secret ingredient in Robitu
ssin I believe I have found it. I got this in addition to the Root Beer Extrac
t I ordered (which was good) and made some cherry soda. The flavor is very med
icinal.
5      B006K2ZZ7K      A1UQRSCLF8GW1T  "Michael D. Bigham ""M. Wassir"" 0
0      5      1350777600      Great taffy      Great taffy at a great price.
There was a wide assortment of yummy taffy. Delivery was very quick. If your a
taffy lover, this is a deal.

C:\Users\shari\Amazon-Fine-Food-Hadoop>
```

First five rows of sample_amazon_reviews.tsv.

Dataset Files After Preprocessing

```
C:\Users\shari\Amazon-Fine-Food-Hadoop>dir dataset
Volume in drive C is Windows
Volume Serial Number is 6207-0771

Directory of C:\Users\shari\Amazon-Fine-Food-Hadoop\dataset

09/12/2026  12:47 AM    <DIR>          .
09/12/2026  12:42 AM    <DIR>          ..
09/12/2026  12:47 AM             299,117,356 amazon_reviews.tsv
09/12/2026  12:47 AM              262 preprocessing_summary.txt
09/12/2026  12:43 AM          300,904,694 Reviews.csv
09/12/2026  12:24 AM          120,062,355 Reviews.csv.zip
09/12/2026  12:47 AM              7,240 sample_amazon_reviews.tsv
               5 File(s)      720,091,907 bytes
               2 Dir(s)  17,547,046,912 bytes free
```

Dataset directory showing raw, compressed, cleaned and sample files.

Preprocessing Verification

The preprocessing script processed 568,454 CSV records, retained 568,452 valid records, removed 2 invalid records, and wrote a 299,117,356-byte TSV file.

```
=====
Input : C:\Users\shari\Amazon-Fine-Food-Hadoop\dataset\Reviews.csv
Output: C:\Users\shari\Amazon-Fine-Food-Hadoop\dataset\amazon_reviews.tsv

Processed: 50,000 | Valid: 49,999 | Invalid: 1
Processed: 100,000 | Valid: 99,998 | Invalid: 2
Processed: 150,000 | Valid: 149,998 | Invalid: 2
Processed: 200,000 | Valid: 199,998 | Invalid: 2
Processed: 250,000 | Valid: 249,998 | Invalid: 2
Processed: 300,000 | Valid: 299,998 | Invalid: 2
Processed: 350,000 | Valid: 349,998 | Invalid: 2
Processed: 400,000 | Valid: 399,998 | Invalid: 2
Processed: 450,000 | Valid: 449,998 | Invalid: 2
Processed: 500,000 | Valid: 499,998 | Invalid: 2
Processed: 550,000 | Valid: 549,998 | Invalid: 2

=====
PREPROCESSING COMPLETED
=====
Total records : 568,454
Valid records : 568,452
Invalid rows : 2
Output size : 299,117,356 bytes

Generated files:
1. amazon_reviews.tsv
2. sample_amazon_reviews.tsv
3. preprocessing_summary.txt

C:\Users\shari\Amazon-Fine-Food-Hadoop>type dataset\preprocessing_summary.txt
Amazon Fine Food Reviews - Preprocessing Summary
=====
Total CSV records : 568,454
Valid records : 568,452
Invalid records : 2
Output file : amazon_reviews.tsv
Output size : 299,117,356 bytes

C:\Users\shari\Amazon-Fine-Food-Hadoop>
```

Verified preprocessing summary from the completed Python script.

HDFS Upload

The cleaned dataset was uploaded to /amazon/input/amazon_reviews.tsv. The HDFS listing reports approximately 285.3 MB.

```
C:\Users\shari\Amazon-Fine-Food-Hadoop>hdfs dfs -put "C:/Users/shari/Amazon-Fine-Food-Hadoop/dataset/amazon_reviews.tsv" /amazon/input/

C:\Users\shari\Amazon-Fine-Food-Hadoop>hdfs dfs -ls -h /amazon/input
Found 1 items
-rw-r--r-- 1 shari supergroup 285.3 M 2026-09-12 01:01 /amazon/input/amazon_reviews.tsv

C:\Users\shari\Amazon-Fine-Food-Hadoop>hdfs dfs -count -h /amazon/input
1 1 285.3 M /amazon/input

C:\Users\shari\Amazon-Fine-Food-Hadoop>hdfs dfs -cat /amazon/input/amazon_reviews.tsv | more
1 B001E4KFG0 A3SGXH7AUHU8GW delmartian 1 1 5 1303862400 Good Quality Dog Food I have bought several of the Vitality
canned dog food products and have found them all to be of good quality. The product looks more like a stew than a processed meat and it smells better. My Labr
ador is finicky and she appreciates this product better than most.
2 B00813GRG4 A1D87F6ZCVE5NK dll pa 0 0 1 1346976000 Not as Advertised "Product arrived labeled as Jumbo Salted Peanu
ts...the peanuts were actually small sized unsalted. Not sure if this was an error or if the vendor intended to represent the product as ""Jumbo""."
3 B000LQOCH0 ABXLMWJIXXAIN "Natalia Corres ""Natalia Corres"" 1 1 4 1219017600 ""Delight"" says it all" "This
is a confection that has been around a few centuries. It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares
and then liberally coated with powdered sugar. And it is a tiny mouthful of heaven. Not too chewy, and very flavorful. I highly recommend this yummy treat. I
f you are familiar with the story of C.S. Lewis' ""The Lion, The Witch, and The Wardrobe"" - this is the treat that seduces Edmund into selling out his Brothe
r and Sisters to the Witch."
4 B000UA0QIQ A395B0RC6FGVXV Karl 3 3 2 1307923200 Cough Medicine If you are looking for the secret ingredient in Robitu
ssin I believe I have found it. I got this in addition to the Root Beer Extract I ordered (which was good) and made some cherry soda. The flavor is very medic
inal.
5 B006K2ZZ7K A1UQRSCLF8GW1T "Michael D. Bigham ""M. Wassir"" 0 0 5 1350777600 Great taffy Great taffy at a great
price. There was a wide assortment of yummy taffy. Delivery was very quick. If your a taffy lover, this is a deal.
6 B006K2ZZ7K ADT0SRK1MG0EU Twoapennything 0 0 4 1342051200 Nice Taffy I got a wild hair for taffy and ordered this f
ive pound bag. The taffy was all very enjoyable with many flavors: watermelon, root beer, melon, peppermint, grape, etc. My only complaint is there was a bit
too much red/black licorice-flavored pieces (just not my particular favorites). Between me, my kids, and my husband, this lasted only two weeks! I would recom
mend this brand of taffy -- it was a delightful treat.
```

HDFS upload, size verification and readable sample from amazon_reviews.tsv.

5. Map Reduce Analysis

Three Java MapReduce analyses were implemented. Each analysis contains a Mapper, Reducer and Driver class. The cleaned HDFS TSV file is used as the input for all three jobs.

5.1 Best Reviewer Based on Number of Reviews

Returns the number of reviews submitted by each UserId and identifies the most active reviewers. The Mapper emits UserId -> 1 for each valid row; the Reducer sums the values for every UserId.

Mapper Class

```
package com.amazon.bestreviewer;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class BestReviewerMapper extends Mapper<LongWritable, Text, Text, IntWritable> {
    private static final IntWritable ONE = new IntWritable(1);
    private final Text userId = new Text();

    @Override
    protected void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {
        String[] fields = value.toString().split("\\t", -1);
        if (fields.length >= 10) {
            String user = fields[2].trim();
            if (!user.isEmpty()) {
                userId.set(user);
                context.write(userId, ONE);
            } else {
                context.getCounter("AMAZON_DATA", "EMPTY_USER_ID").increment(1);
            }
        } else {
            context.getCounter("AMAZON_DATA", "MALFORMED_ROWS").increment(1);
        }
    }
}
```

BestReviewerMapper.java

Reducer Class

```
package com.amazon.bestreviewer;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class BestReviewerReducer extends Reducer<Text, IntWritable, Text, IntWritable> {
    @Override
    protected void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {
        int totalReviews = 0;
        for (IntWritable value : values) totalReviews += value.get();
        context.write(key, new IntWritable(totalReviews));
    }
}
```

BestReviewerReducer.java

Driver Class

```
package com.amazon.bestreviewer;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class BestReviewerDriver {
    public static void main(String[] args) throws Exception {
        if (args.length != 2) System.exit(2);
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Best Reviewer Based on Number of Reviews");
        job.setJarByClass(BestReviewerDriver.class);
        job.setMapperClass(BestReviewerMapper.class);
        job.setReducerClass(BestReviewerReducer.class);
        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(IntWritable.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

BestReviewerDriver.java

Execution

```
C:\Users\shari\Amazon-Fine-Food-Hadoop>hdfs dfs -rm -r -f /amazon/mapreduce_output/best_reviewer

C:\Users\shari\Amazon-Fine-Food-Hadoop>hadoop jar jar\best-reviewer.jar com.amazon.bestreviewer.BestReviewerDriver /amazon/input/amazon_reviews.tsv /amazon/mapreduce_output/best_reviewer
2026-09-12 01:08:07,903 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 01:08:08,719 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-12 01:08:08,747 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/shari/.staging/job_1789151467584_0001
2026-09-12 01:08:08,990 INFO input.FileInputFormat: Total input files to process : 1
2026-09-12 01:08:09,095 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-12 01:08:09,233 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789151467584_0001
2026-09-12 01:08:09,235 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-12 01:08:09,400 INFO conf.Configuration: resource-types.xml not found
2026-09-12 01:08:09,401 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-12 01:08:10,327 INFO impl.YarnClientImpl: Submitted application application_1789151467584_0001
2026-09-12 01:08:10,363 INFO mapreduce.Job: The url to track the job: http://LAPTOP-4HNA6QNE:8088/proxy/application_1789151467584_0001/
2026-09-12 01:08:10,364 INFO mapreduce.Job: Running job: job_1789151467584_0001
2026-09-12 01:08:25,736 INFO mapreduce.Job: Job job_1789151467584_0001 running in uber mode : false
2026-09-12 01:08:25,739 INFO mapreduce.Job: map 0% reduce 0%
2026-09-12 01:08:36,100 INFO mapreduce.Job: map 33% reduce 0%
2026-09-12 01:08:38,128 INFO mapreduce.Job: map 100% reduce 0%
2026-09-12 01:08:46,234 INFO mapreduce.Job: map 100% reduce 100%
2026-09-12 01:08:47,261 INFO mapreduce.Job: Job job_1789151467584_0001 completed successfully
2026-09-12 01:08:47,411 INFO mapreduce.Job: Counters: 55
    File System Counters
        FILE: Number of bytes read=11790174
        FILE: Number of bytes written=24536913
        FILE: Number of read operations=0
        FILE: Number of large read operations=0
        FILE: Number of write operations=0
        HDFS: Number of bytes read=299125902
        HDFS: Number of bytes written=4294864
        HDFS: Number of read operations=14
        HDFS: Number of large read operations=0
        HDFS: Number of write operations=2
        HDFS: Number of bytes read erasure-coded=0
```

MRI Hadoop job completed successfully with map 100% and reduce 100%.

Result

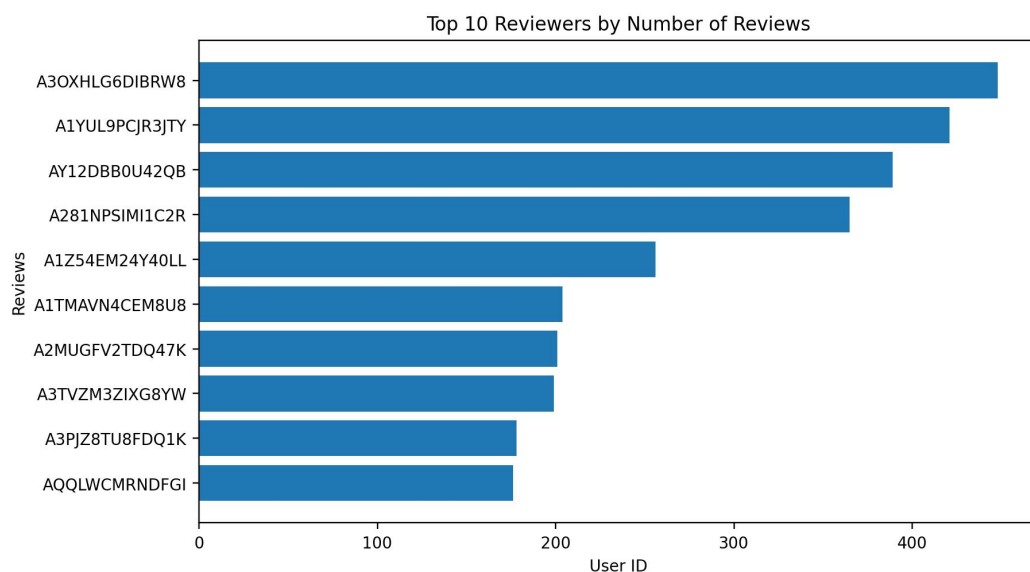
```
C:\Users\shari\Amazon-Fine-Food-Hadoop>powershell -NoProfile -Command "Get-Content 'results\mapreduce\mr1_best_reviewer.txt' | ForEach-Object { $p = $_ -split '\t'; if ($p.Count -ge 2) { [PSCustomObject]@{UserId=$p[0]; Reviews=[int]$p[1]} } } | Sort-Object Reviews -Descending | Select-Object -First 10 | Format-Table -AutoSize"

UserId      Reviews
-----
A3OXHLG6DIBRW8    448
A1YUL9PCJR3JTY    421
AY12DBB0U42QB     389
A281NPSIMI1C2R     365
A1Z54EM24Y40LL     256
A1TMAVN4CEM8U8     204
A2MUGFV2TDQ47K     201
A3TVZM3ZIXG8YW     199
A3PJZ8TU8FDQ1K     178
AQQLWCMRNDFGI     176
```

Top reviewers sorted by review count using the saved MapReduce result.

User ID	Reviews
A3OXHLG6DIBRW8	448
A1YUL9PCJR3JTY	421
AY12DBB0U42QB	389
A281NPSIMI1C2R	365
A1Z54EM24Y40LL	256
A1TMAVN4CEM8U8	204
A2MUGFV2TDQ47K	201
A3TVZM3ZIXG8YW	199
A3PJZ8TU8FDQ1K	178
AQQLWCMRNDFGI	176

Interpretation: A3OXHLG6DIBRW8 was the most active reviewer with 448 reviews, followed by A1YUL9PCJR3JTY with 421 and AY12DBB0U42QB with 389 reviews.



Top 10 reviewers by number of submitted reviews.

5.2 Total Amazon Reviews by Rating Score

Counts the number of valid reviews for rating scores 1 through 5. The Mapper emits Score -> 1 and the Reducer sums each score group.

Mapper Class

```
package com.amazon.ratingcount;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class RatingCountMapper extends Mapper<LongWritable, Text, Text, IntWritable> {
    private static final IntWritable ONE = new IntWritable(1);
    private final Text scoreKey = new Text();
    @Override
    protected void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
        String[] fields = value.toString().split("\\t", -1);
        if (fields.length >= 10) {
            try {
                int score = Integer.parseInt(fields[6].trim());
                if (score >= 1 && score <= 5) { scoreKey.set(String.valueOf(score)); context.write(scoreKey, ONE); }
                else context.getCounter("AMAZON_DATA", "INVALID_SCORE").increment(1);
            } catch (NumberFormatException e) { context.getCounter("AMAZON_DATA", "INVALID_SCORE").increment(1); }
            else context.getCounter("AMAZON_DATA", "MALFORMED_ROWS").increment(1);
        }
    }
}
```

RatingCountMapper.java

Reducer Class

```
package com.amazon.ratingcount;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class RatingCountReducer extends Reducer<Text, IntWritable, Text, IntWritable> {
    @Override
    protected void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {
        int totalReviews = 0;
        for (IntWritable value : values) totalReviews += value.get();
        context.write(key, new IntWritable(totalReviews));
    }
}
```

RatingCountReducer.java

Driver Class

```
package com.amazon.ratingcount;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class RatingCountDriver {
    public static void main(String[] args) throws Exception {
        if (args.length != 2) System.exit(2);
        Job job = Job.getInstance(new Configuration(), "Total Amazon Reviews by Rating Score");
        job.setJarByClass(RatingCountDriver.class);
        job.setMapperClass(RatingCountMapper.class);
        job.setReducerClass(RatingCountReducer.class);
        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(IntWritable.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

RatingCountDriver.java

Execution

```
C:\Users\shari\Amazon-Fine-Food-Hadoop>hdfs dfs -rm -r -f /amazon/mapreduce_output/rating_count

C:\Users\shari\Amazon-Fine-Food-Hadoop>hadoop jar jar\rating-count.jar com.amazon.ratingcount.RatingCountDriver /amazon/input/amazon_reviews.tsv /amazon/mapreduce_output/rating_count
2026-09-12 01:23:30,596 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 01:23:31,386 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-12 01:23:31,408 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/shari/.staging/job_1789151467584_0002
2026-09-12 01:23:31,669 INFO input.FileInputFormat: Total input files to process : 1
2026-09-12 01:23:31,784 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-12 01:23:31,940 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789151467584_0002
2026-09-12 01:23:31,941 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-12 01:23:32,134 INFO conf.Configuration: resource-types.xml not found
2026-09-12 01:23:32,134 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-12 01:23:32,216 INFO impl.YarnClientImpl: Submitted application application_1789151467584_0002
2026-09-12 01:23:32,246 INFO mapreduce.Job: The url to track the job: http://LAPTOP-4HNA6QNE:8088/proxy/application_1789151467584_0002/
2026-09-12 01:23:32,247 INFO mapreduce.Job: Running job: job_1789151467584_0002
2026-09-12 01:23:43,533 INFO mapreduce.Job: Job job_1789151467584_0002 running in uber mode : false
2026-09-12 01:23:43,535 INFO mapreduce.Job: map 0% reduce 0%
2026-09-12 01:23:52,824 INFO mapreduce.Job: map 33% reduce 0%
2026-09-12 01:23:54,849 INFO mapreduce.Job: map 100% reduce 0%
2026-09-12 01:24:02,028 INFO mapreduce.Job: map 100% reduce 100%
2026-09-12 01:24:03,057 INFO mapreduce.Job: Job job_1789151467584_0002 completed successfully
2026-09-12 01:24:03,265 INFO mapreduce.Job: Counters: 54
  File System Counters
    FILE: Number of bytes read=4547622
    FILE: Number of bytes written=10051773
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=299125902
    HDFS: Number of bytes written=41
    HDFS: Number of read operations=14
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
```

MR2 Hadoop job completed successfully.

Result

```
C:\Users\shari\Amazon-Fine-Food-Hadoop>hdfs dfs -ls /amazon/mapreduce_output/rating_count
Found 2 items
-rw-r--r--  1 shari supergroup          0 2026-09-12 01:24 /amazon/mapreduce_output/rating_count/_SUCCESS
-rw-r--r--  1 shari supergroup       41 2026-09-12 01:24 /amazon/mapreduce_output/rating_count/part-r-00000

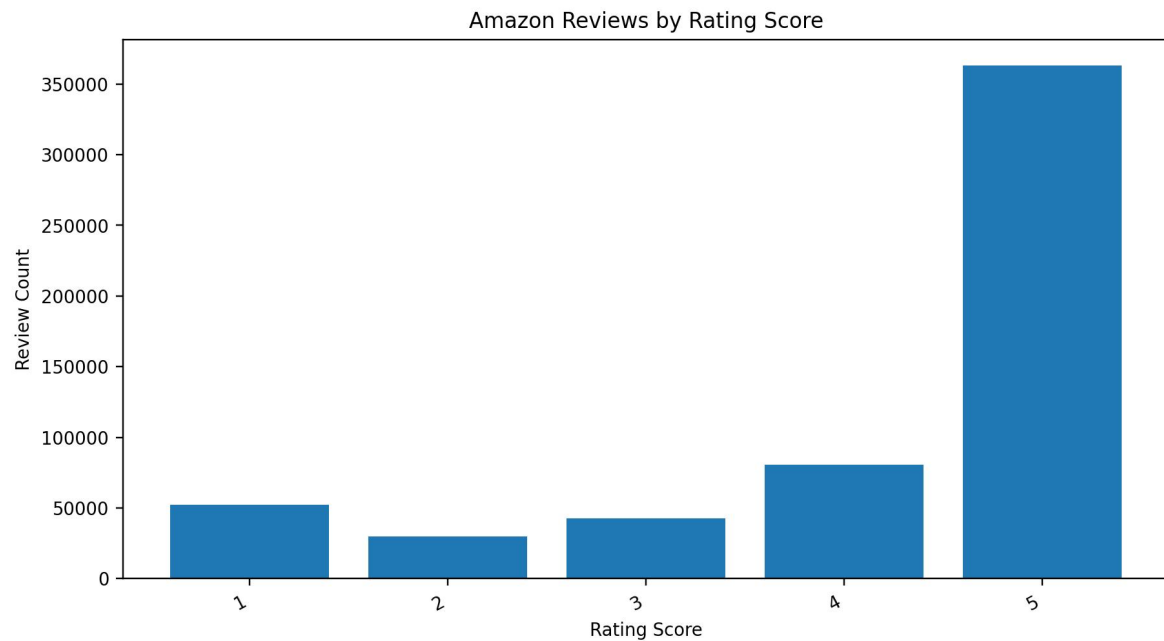
C:\Users\shari\Amazon-Fine-Food-Hadoop>hdfs dfs -cat /amazon/mapreduce_output/rating_count/part-r-00000
1      52268
2      29769
3      42640
4      80654
5     363121

C:\Users\shari\Amazon-Fine-Food-Hadoop>
```

HDFS output for review counts by rating score.

Rating Score	Review Count
1	52,268
2	29,769
3	42,640
4	80,654
5	363,121

Interpretation: Score 5 is dominant with 363,121 reviews, followed by score 4 with 80,654. Score 2 is the smallest group with 29,769 reviews. The total of all five groups is 568,452, exactly matching the valid-record count from preprocessing.



Distribution of valid Amazon reviews by rating score.

5.3 Total Helpful Votes Based on Product ID

Calculates the total helpful votes received by each product. The Mapper emits ProductId -> HelpfulnessNumerator, and the Reducer sums the values for each product.

Mapper Class

```
package com.amazon.producthelpfulness;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class ProductHelpfulnessMapper extends Mapper<LongWritable, Text, Text, IntWritable> {
    private final Text productId = new Text();
    private final IntWritable helpfulVotes = new IntWritable();
    @Override
    protected void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
        String[] fields = value.toString().split("\\t", -1);
        if (fields.length >= 10) {
            String product = fields[1].trim();
            try {
                int helpful = Integer.parseInt(fields[4].trim());
                if (!product.isEmpty() && helpful >= 0) { productId.set(product); helpfulVotes.set(helpful); context.write(productId, helpfulVotes); }
                else context.getCounter("AMAZON_DATA", "INVALID_PRODUCT_OR_HELPFULNESS").increment(1);
            } catch (NumberFormatException e) { context.getCounter("AMAZON_DATA", "INVALID_HELPFULNESS").increment(1); }
            else context.getCounter("AMAZON_DATA", "MALFORMED_ROWS").increment(1);
        }
    }
}
```

ProductHelpfulnessMapper.java

Reducer Class

```
package com.amazon.producthelpfulness;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class ProductHelpfulnessReducer extends Reducer<Text, IntWritable, Text, LongWritable> {
    @Override
    protected void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {
        long totalHelpfulVotes = 0;
        for (IntWritable value : values) totalHelpfulVotes += value.get();
        context.write(key, new LongWritable(totalHelpfulVotes));
    }
}
```

ProductHelpfulnessReducer.java

Driver Class

```
package com.amazon.producthelpfulness;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class ProductHelpfulnessDriver {
    public static void main(String[] args) throws Exception {
        if (args.length != 2) System.exit(2);
        Job job = Job.getInstance(new Configuration(), "Total Helpful Votes Based on Product ID");
        job.setJarByClass(ProductHelpfulnessDriver.class);
        job.setMapperClass(ProductHelpfulnessMapper.class);
        job.setReducerClass(ProductHelpfulnessReducer.class);
        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(IntWritable.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(LongWritable.class);
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

ProductHelpfulnessDriver.java

Execution

```
C:\Users\shari\Amazon-Fine-Food-Hadoop>hadoop jar jar\product-helpfulness.jar com.amazon.producthelpfulness.ProductHelpfulnessDriver /amazon/input/amazon_reviews.tsv /amazon/output/product_helpfulness
2026-09-12 01:30:36,721 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 01:30:37,473 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-12 01:30:37,493 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/shari/.staging/job_1789151467584_0003
2026-09-12 01:30:37,714 INFO input.FileInputFormat: Total input files to process : 1
2026-09-12 01:30:37,814 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-12 01:30:37,962 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789151467584_0003
2026-09-12 01:30:37,963 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-12 01:30:38,160 INFO conf.Configuration: resource-types.xml not found
2026-09-12 01:30:38,160 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-12 01:30:38,254 INFO impl.YarnClientImpl: Submitted application application_1789151467584_0003
2026-09-12 01:30:38,297 INFO mapreduce.Job: The url to track the job: http://LAPTOP-4HNA6QNE:8088/proxy/application_1789151467584_0003/
2026-09-12 01:30:38,298 INFO mapreduce.Job: Running job: job_1789151467584_0003
2026-09-12 01:30:48,607 INFO mapreduce.Job: Job job_1789151467584_0003 running in uber mode : false
2026-09-12 01:30:48,608 INFO mapreduce.Job: map 0% reduce 0%
2026-09-12 01:30:57,895 INFO mapreduce.Job: map 33% reduce 0%
2026-09-12 01:30:59,923 INFO mapreduce.Job: map 67% reduce 0%
2026-09-12 01:31:00,938 INFO mapreduce.Job: map 100% reduce 0%
2026-09-12 01:31:08,037 INFO mapreduce.Job: map 100% reduce 100%
2026-09-12 01:31:08,045 INFO mapreduce.Job: Job job_1789151467584_0003 completed successfully
2026-09-12 01:31:08,169 INFO mapreduce.Job: Counters: 55
File System Counters
  FILE: Number of bytes read=9663690
  FILE: Number of bytes written=20284065
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=299125902
  HDFS: Number of bytes written=982088
  HDFS: Number of read operations=14
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
```

MR3 Hadoop job completed successfully.

Result

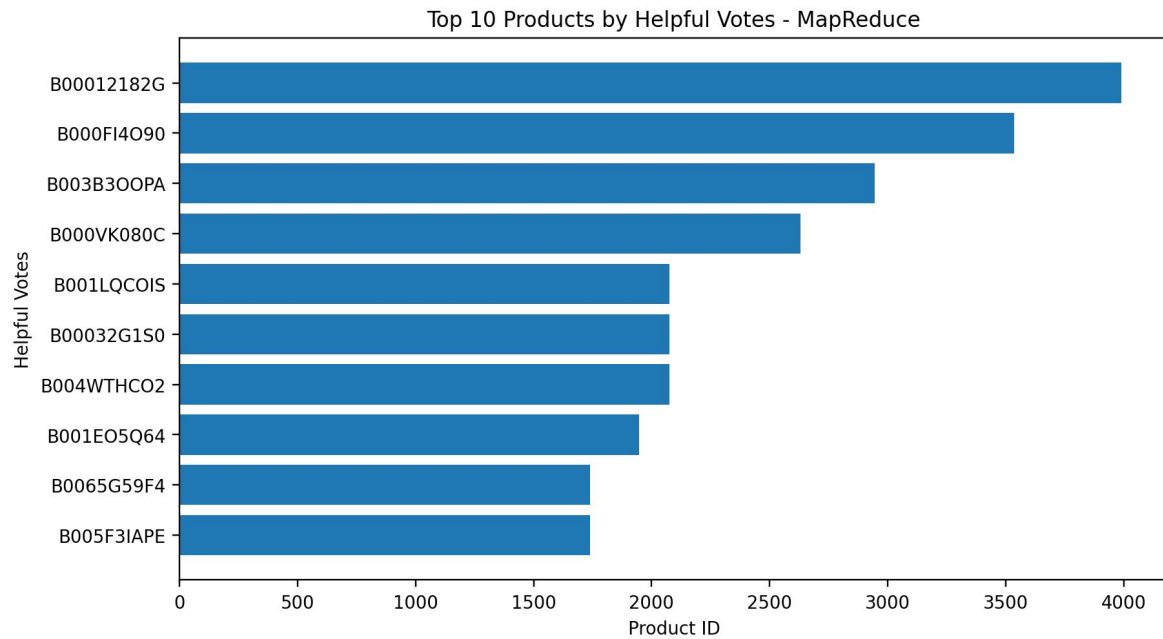
```
C:\Users\shari\Amazon-Fine-Food-Hadoop>powershell -NoProfile -Command "Get-Content 'results\mapreduce\mr3_product_helpfulness.txt' | ForEach-Object { $p = $_ -split '\t'; if ($p.Count -ge 2) { [PSCustomObject]@{ProductId=$p[0]; HelpfulVotes=[long]$p[1]} } } | Sort-Object HelpfulVotes -Descending | Select-Object -First 10 | Format-Table -AutoSize"
```

ProductId	HelpfulVotes
B00012182G	3989
B000FI4O90	3535
B003B3OOPA	2944
B000VK080C	2630
B001LQCOIS	2074
B00032G1S0	2074
B004WTHCO2	2074
B001EO5Q64	1947
B0065G59F4	1739
B005F3IAPE	1739

Top products sorted by total helpful votes from the MR3 result file.

Product ID	Helpful Votes
B00012182G	3,989
B000FI4O90	3,535
B003B3OOPA	2,944
B000VK080C	2,630
B001LQCOIS	2,074
B00032G1S0	2,074
B004WTHCO2	2,074
B001EO5Q64	1,947
B0065G59F4	1,739
B005F3IAPE	1,739

Interpretation: B00012182G received the highest total helpful-vote count with 3,989, followed by B000FI4O90 with 3,535 and B003B3OOPA with 2,944. Several products are tied at 1,739 helpful votes, so the exact product IDs occupying the final tied positions can vary when no secondary sort key is defined.



Top 10 products by total helpful votes from MapReduce.

6. Pig Amazon Fine Food Reviews Data Analysis

Apache Pig was used for five required analyses on the same cleaned HDFS dataset. The scripts use PigStorage(\t), FILTER, GROUP, FOREACH, COUNT/SUM, ORDER, LIMIT and STORE operations.

6.1 Top 5 Products by Number of Reviews

Returns the five Product IDs containing the largest number of valid reviews.

Script

```
reviews = LOAD '/amazon/input/amazon_reviews.tsv' USING PigStorage('\t') AS
(id:long, productid:chararray, userid:chararray, profilename:chararray,
helpful_num:int, helpful_den:int, score:int, review_time:long,
summary:chararray, review_text:chararray);
valid = FILTER reviews BY productid IS NOT NULL AND productid != "";
grouped = GROUP valid BY productid;
counts = FOREACH grouped GENERATE group AS productid, COUNT(valid) AS review_count;
sorted = ORDER counts BY review_count DESC;
top5 = LIMIT sorted 5;
STORE top5 INTO '/amazon/pig_output/top5_products' USING PigStorage("");
```

top5Products.pig

Execution and Result Evidence

```
C:\Users\shari\Amazon-Fine-Food-Hadoop>hdfs dfs -cat /amazon/pig_output/top5_products/part-r-00000 > results\pig\pig1_top5_products.txt

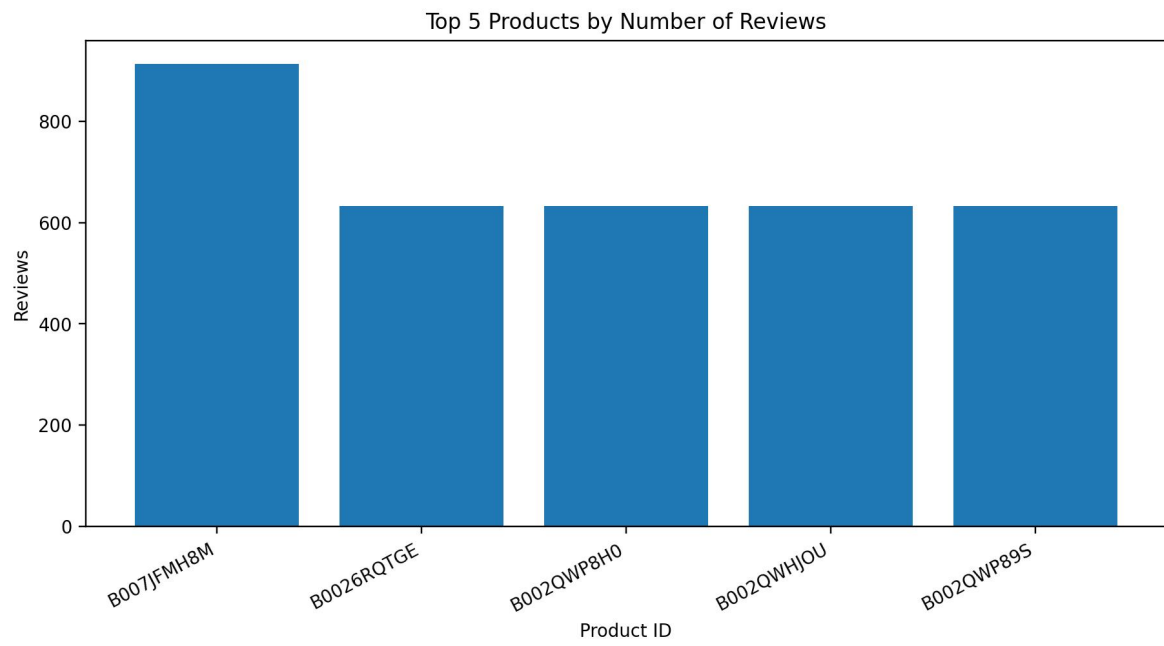
C:\Users\shari\Amazon-Fine-Food-Hadoop>type results\pig\pig1_top5_products.txt
B007JFMH8M|913
B0026RQTGE|632
B002QWP8H0|632
B002QWHJOU|632
B002QWP89S|632

C:\Users\shari\Amazon-Fine-Food-Hadoop>
```

Pig 1 HDFS result verification and local result-file creation.

Product ID	Review Count
B007JFMH8M	913
B0026RQTGE	632
B002QWP8H0	632
B002QWHJOU	632
B002QWP89S	632

Interpretation: Product B007JFMH8M had the highest number of reviews with 913. The other four products shown each had 632 reviews.



Top 5 products by number of reviews.

6.2 Top 10 Most Helpful Reviews

Ranks individual reviews by HelpfulnessNumerator and returns the ten reviews with the largest helpful-vote counts.

Script

```
reviews = LOAD '/amazon/input/amazon_reviews.tsv' USING PigStorage('\t') AS
(id:long, productid:chararray, userid:chararray, profilename:chararray,
helpful_num:int, helpful_den:int, score:int, review_time:long,
summary:chararray, review_text:chararray);
valid = FILTER reviews BY id IS NOT NULL AND productid IS NOT NULL
AND helpful_num IS NOT NULL AND score IS NOT NULL;
ordered = ORDER valid BY helpful_num DESC;
top10 = LIMIT ordered 10;
result = FOREACH top10 GENERATE id, productid, score, helpful_num;
STORE result INTO '/amazon/pig_output/top10_helpful_reviews' USING PigStorage('|');
```

top10HelpfulReviews.pig

Execution

```
2026-09-12 02:03:42,705 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:03:42,711 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:03:42,754 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:03:42,761 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:03:42,809 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:03:42,814 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:03:42,866 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:03:42,872 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:03:42,923 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:03:42,928 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:03:42,969 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:03:42,974 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:03:43,019 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:03:43,024 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:03:43,065 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:03:43,071 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:03:43,107 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:03:43,112 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:03:43,151 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-12 02:03:43,204 [main] INFO org.apache.pig.Main - Pig script completed in 2 minutes, 9 seconds and 671 milliseconds (129671 ms)
)
C:\Users\shari\Amazon-Fine-Food-Hadoop>
```

Pig 2 execution completed successfully.

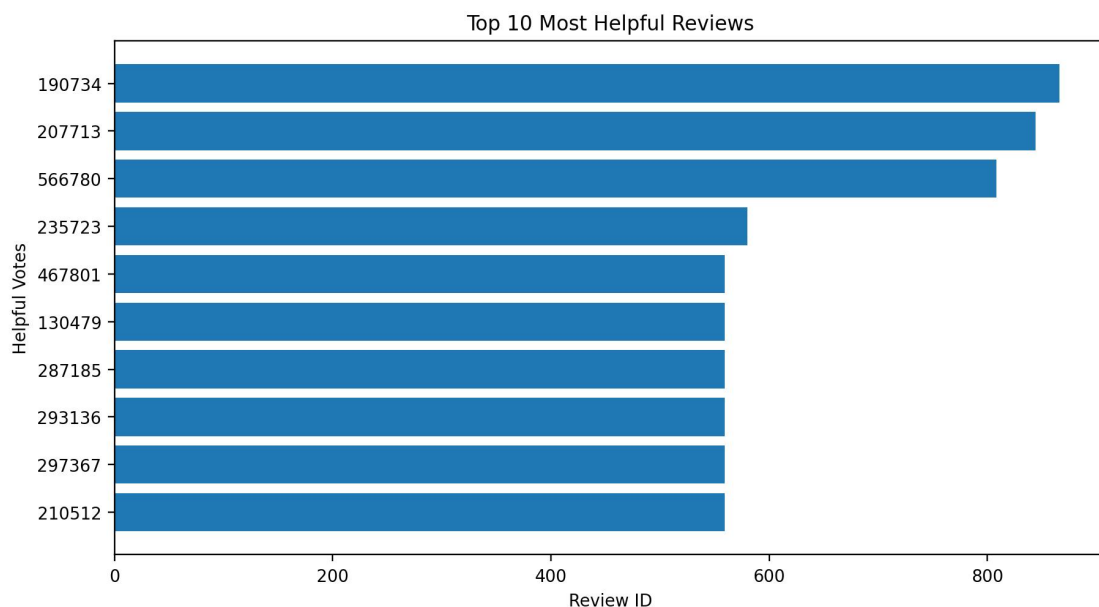
Result

```
C:\Users\shari\Amazon-Fine-Food-Hadoop>type results\pig\pig2_top10_helpful_reviews.txt
190734|B000FI4O90|5|866
207713|B00012182G|3|844
566780|B001PQTYN2|5|808
235723|B001F10XUU|1|580
467801|B0065G59F4|5|559
130479|B005F3IAPE|5|559
287185|B009K2BBT8|5|559
293136|B008VQ01Y0|5|559
297367|B0069GOKGE|5|559
210512|B005FKRTO0|5|559
```

Top 10 most helpful individual reviews.

Review ID	Product ID	Score	Helpful Votes
190734	B000FI4O90	5	866
207713	B00012182G	3	844
566780	B001PQTYN2	5	808
235723	B001F10XUU	1	580
467801	B0065G59F4	5	559
130479	B005F3IAPE	5	559
287185	B009K2BBT8	5	559
293136	B008VQ01Y0	5	559
297367	B0069GOKGE	5	559
210512	B005FKRTO0	5	559

Interpretation: Review 190734 for product B000FI4O90 ranked first with 866 helpful votes and score 5. Review 207713 followed with 844 votes, and Review 566780 had 808 votes.



Top 10 most helpful reviews.

6.3 Top 10 Helpful Reviews by Rating Score

Groups reviews by rating score and ranks the ten most helpful reviews inside each score group. The output therefore contains 50 rows: ten for each score from 1 to 5.

Script

```
reviews = LOAD '/amazon/input/amazon_reviews.tsv' USING PigStorage('\t') AS
(id:long, productid:chararray, userid:chararray, profilename:chararray,
 helpful_num:int, helpful_den:int, score:int, review_time:long,
 summary:chararray, review_text:chararray);
valid = FILTER reviews BY id IS NOT NULL AND productid IS NOT NULL
      AND helpful_num IS NOT NULL AND score IS NOT NULL AND score >= 1 AND score <= 5;
clean = FOREACH valid GENERATE id, productid, score, helpful_num;
grouped = GROUP clean BY score;
ranked = FOREACH grouped {
  ordered = ORDER clean BY helpful_num DESC, id ASC;
  top10 = LIMIT ordered 10;
  GENERATE FLATTEN(top10);
};
STORE ranked INTO '/amazon/pig_output/top10_helpful_by_score' USING PigStorage('|');
```

top10HelpfulReviewsByScore.pig

Execution

```
Input(s):
Successfully read 568452 records (299126688 bytes) from: "/amazon/input/amazon_reviews.tsv"

Output(s):
Successfully stored 50 records (1194 bytes) in: "/amazon/pig_output/top10_helpful_by_score"

Counters:
Total records written : 50
Total bytes written : 1194
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789151467584_0015

2026-09-12 02:08:05,204 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:08:05,212 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:08:05,270 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:08:05,277 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:08:05,311 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:08:05,320 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:08:05,360 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-12 02:08:05,411 [main] INFO org.apache.pig.Main - Pig script completed in 46 seconds and 56 milliseconds (46056 ms)

C:\Users\shari\Amazon-Fine-Food-Hadoop>
```

Pig 3 execution successfully read 568,452 records and stored 50 ranked rows.

Result Evidence

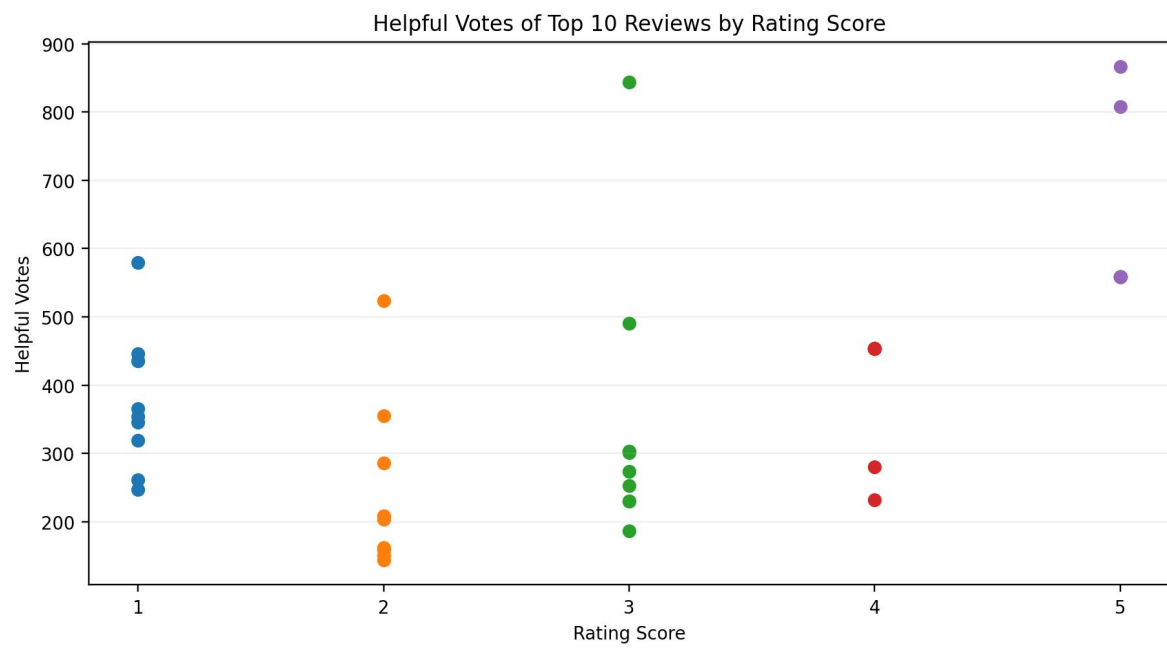
```
C:\Users\shari\Amazon-Fine-Food-Hadoop>powershell -NoProfile -Command "Get-Content 'results\pig\pig3_top10_helpful_by_score.txt' -TotalCount 20"
235723|B001F10XUU|1|580
260703|B0099HD3YA|1|446
241669|B0081XPTBS|1|436
358114|B0032CJPOK|1|436
155533|B000FFLHU2|1|366
279313|B005NYXE1K|1|354
268062|B00032G1S0|1|346
260705|B0099HD3YA|1|319
260704|B0099HD3YA|1|262
155532|B000FFLHU2|1|247
190735|B000FI4O90|2|524
226506|B000A3I3BA|2|355
67263|B004JRXZKM|2|286
373782|B004WTHCO2|2|209
433681|B001LQCOIS|2|209
458304|B00158U8DU|2|204
122088|B000ORSIJ4|2|162
438433|B00015UC48|2|159
408264|B005NYXE24|2|151
245665|B0040AZ36I|2|144

C:\Users\shari\Amazon-Fine-Food-Hadoop>
```

Beginning of pig3_top10_helpful_by_score.txt, showing score-1 and score-2 rankings.

Score	Top Review ID	Product ID	Helpful Votes
1	235723	B001F10XUU	580
2	190735	B000FI4O90	524
3	207713	B00012182G	844
4	130827	B001CHFUDC	454
5	190734	B000FI4O90	866

Interpretation: The leading review for score 1 had 580 helpful votes; score 2 was led by 524; score 3 by 844; several score-4 reviews tied at 454; and score 5 was led by 866 helpful votes. The screenshot shows the start of the 50-row output, while the script itself stored all 50 rows.



Helpful-vote distribution of the Top 10 reviews within each rating score.

6.4 Top 10 Products by Total Helpful Votes

Groups all reviews by ProductId, sums HelpfulnessNumerator and returns the ten products with the largest total helpful-vote counts.

Script

```
reviews = LOAD 'amazon/input/amazon_reviews.tsv' USING PigStorage('\t') AS
(id:long, productid:chararray, userid:chararray, profilename:chararray,
helpful_num:int, helpful_den:int, score:int, review_time:long,
summary:chararray, review_text:chararray);
valid = FILTER reviews BY productid IS NOT NULL AND productid != "
AND helpful_num IS NOT NULL AND helpful_num >= 0;
grouped = GROUP valid BY productid;
totals = FOREACH grouped GENERATE group AS productid,
SUM(valid.helpful_num) AS total_helpful_votes;
ordered = ORDER totals BY total_helpful_votes DESC;
top10 = LIMIT ordered 10;
STORE top10 INTO 'amazon/pig_output/top10_products_helpful' USING PigStorage('');
```

top10ProductsByHelpfulVotes.pig

Execution

```
2026-09-12 02:13:47,489 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:13:47,492 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplications
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:13:47,527 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:13:47,531 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplications
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:13:47,566 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:13:47,572 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplications
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:13:47,614 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:13:47,620 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplications
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:13:47,672 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:13:47,678 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplications
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:13:47,715 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:13:47,720 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplications
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:13:47,765 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:13:47,771 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplications
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:13:47,807 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:13:47,812 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplications
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:13:47,854 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:13:47,858 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplications
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:13:47,891 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-12 02:13:47,940 [main] INFO org.apache.pig.Main - Pig script completed in 2 minutes, 36 seconds and 445 milliseconds (156445 m
s)
C:\Users\shari\Amazon-Fine-Food-Hadoop>
```

Pig 4 execution completed successfully.

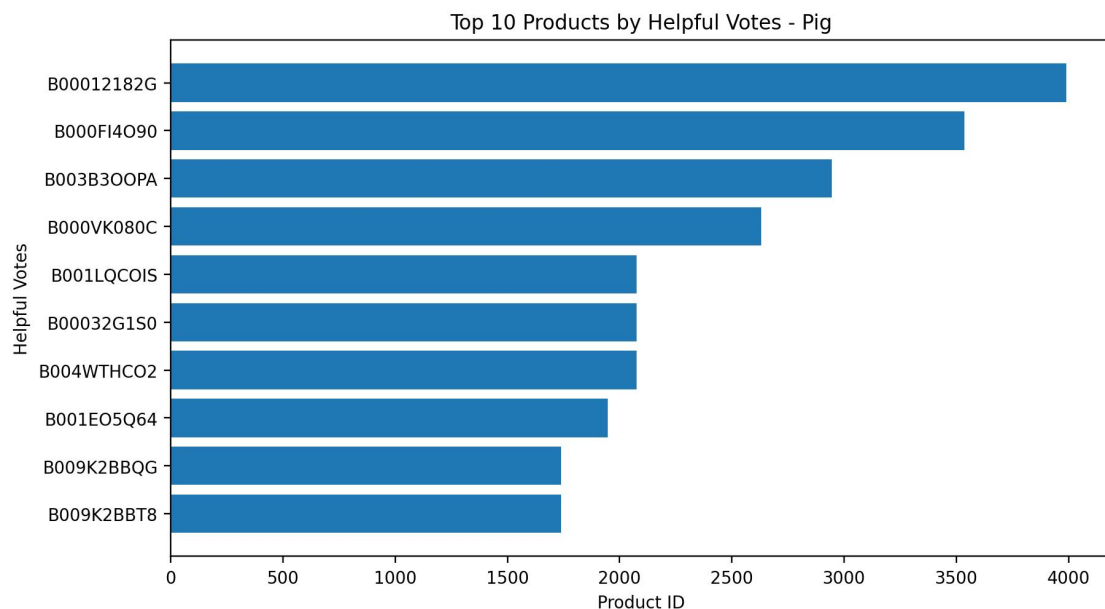
Result

```
C:\Users\shari\Amazon-Fine-Food-Hadoop>type results\pig\pig4_top10_products_helpful.txt
B00012182G| 3989
B000FI4O90| 3535
B003B3OOPA| 2944
B000VK080C| 2630
B001LQCOIS| 2074
B00032G1S0| 2074
B004WTHCO2| 2074
B001EO5Q64| 1947
B009K2BBQG| 1739
B009K2BBT8| 1739
C:\Users\shari\Amazon-Fine-Food-Hadoop>
```

Pig 4 Top 10 product helpfulness result.

Product ID	Helpful Votes
B00012182G	3,989
B000FI4O90	3,535
B003B3OOPA	2,944
B000VK080C	2,630
B001LQCOIS	2,074
B00032G1S0	2,074
B004WTHCO2	2,074
B001EO5Q64	1,947
B009K2BBQG	1,739
B009K2BBT8	1,739

Interpretation: The first eight ranked results match the MapReduce aggregation exactly. At 1,739 helpful votes there are multiple tied products; because the MapReduce and Pig rankings do not define the same secondary tie-break, different tied Product IDs may appear in the last positions while the helpful-vote total remains consistent.



6.5 Top 10 Products by Helpful Votes for Each Rating Score

Groups reviews by the composite key (Score, ProductId), sums helpful votes for each product inside each score group, and returns the ten leading products for every rating score.

Script

```
reviews = LOAD '/amazon/input/amazon_reviews.tsv' USING PigStorage('\t') AS
(id:long, productid:chararray, userid:chararray, profilename:chararray,
 helpful_num:int, helpful_den:int, score:int, review_time:long,
 summary:chararray, review_text:chararray);
valid = FILTER reviews BY productid IS NOT NULL AND productid != ""
      AND helpful_num IS NOT NULL AND helpful_num >= 0
      AND score IS NOT NULL AND score >= 1 AND score <= 5;
product_score_group = GROUP valid BY (score, productid);
product_totals = FOREACH product_score_group GENERATE
  group.score AS score, group.productid AS productid,
  SUM(valid.helpful_num) AS total_helpful_votes;
score_group = GROUP product_totals BY score;
ranked = FOREACH score_group {
  ordered = ORDER product_totals BY total_helpful_votes DESC, productid ASC;
  top10 = LIMIT ordered 10;
  GENERATE FLATTEN(top10);
};
STORE ranked INTO '/amazon/pig_output/top10_products_helpful_by_score' USING PigStorage('|');
```

top10ProductsByHelpfulVotesByScore.pig

Execution

```
Counters:
Total records written : 50
Total bytes written : 862
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789151467584_0020 -> job_1789151467584_0021,
job_1789151467584_0021

2026-09-12 02:19:02,164 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:19:02,167 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:19:02,186 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:19:02,189 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:19:02,207 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:19:02,209 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:19:02,229 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:19:02,231 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:19:02,247 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:19:02,249 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:19:02,270 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 02:19:02,275 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationS
tatus=SUCCEEDED. Redirecting to job history server
2026-09-12 02:19:02,295 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-12 02:19:02,313 [main] INFO org.apache.pig.Main - Pig script completed in 59 seconds and 817 milliseconds (59817 ms)
```

Pig 5 execution completed successfully and stored 50 rows.

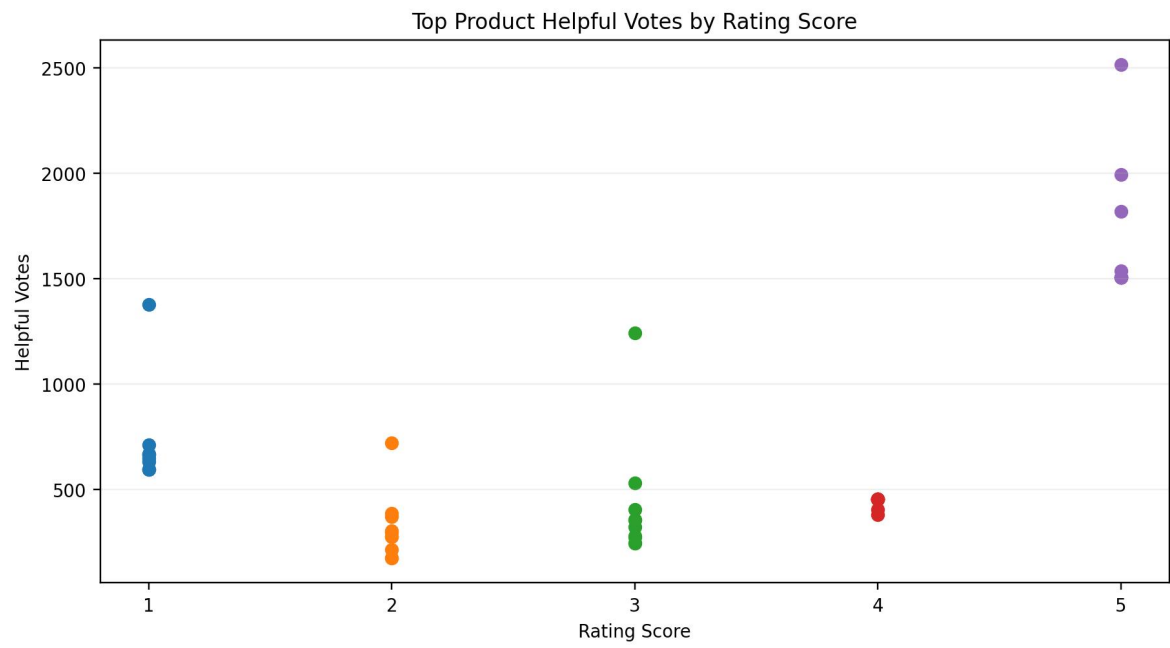
Result Evidence

```
C:\Users\shari\Amazon-Fine-Food-Hadoop>type results\pig\pig5_top10_products_helpful_by_score.txt
1|B0099HD3YA|1377
1|B000FFLHU2|712
1|B00507A02Q|671
1|B0032CJPOK|666
1|B0081XPTBS|666
1|B00012182G|648
1|B001F10XUU|634
1|B001410U50|597
1|B001410X52|597
1|B00141UC9I|597
2|B000FI4090|721
2|B005NYXE24|386
2|B000A3I3BA|371
2|B004JRXZKM|306
2|B00012182G|299
2|B001LQCOIS|277
2|B004WTHCO2|277
2|B00158U8DU|217
2|B0016BAB42|176
2|B000FI2VHM|175
3|B00012182G|1242
3|B000UUWECC|532
3|B002HFPW6I|405
3|B001LQCOIS|358
3|B004WTHCO2|358
3|B001D0KG4Q|324
3|B000FI4090|279
3|B00032G1S0|274
```

Beginning of pig5_top10_products_helpful_by_score.txt, showing score-1, score-2 and part of score-3 rankings.

Score	Leading Product ID	Helpful Votes
1	B0099HD3YA	1377
2	B000FI4090	721
3	B00012182G	1242
4	B001CHFUDC	454
5	B003B3OOPA	2515

Interpretation: B0099HD3YA leads score 1 with 1,377 helpful votes; B000FI4090 leads score 2 with 721; B00012182G leads score 3 with 1,242; several score-4 products tie at 454; and B003B3OOPA leads score 5 with 2,515. The complete Pig output contains 50 rows in total.



Top product helpful-vote distributions by rating score.

7. Conclusion

This project demonstrates a complete Hadoop-based workflow for Amazon Fine Food Reviews. The original CSV was cleaned into a reliable TSV file, stored in HDFS, processed with three Java MapReduce jobs, and analyzed with five Apache Pig scripts. The preprocessing step retained 568,452 valid records from 568,454 source reviews. MapReduce identified the most active reviewer, summarized the rating distribution, and aggregated helpful votes by product. Pig reproduced important aggregation results and added top-product, top-review and score-wise ranking analyses.

Key findings include A3OXHLG6DIBRW8 as the most active reviewer with 448 reviews, score 5 as the dominant rating with 363,121 reviews, B00012182G as the product with the largest total helpful-vote count at 3,989, and B007JFMH8M as the product with the highest review count among the Pig Top 5 result at 913. The screenshots and result files included in this report provide direct execution evidence for the completed Hadoop workflow.

8. References

1. Amazon Fine Food Reviews Dataset, Kaggle: <https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews>
2. Apache Hadoop Documentation: <https://hadoop.apache.org/>
3. Apache Pig Documentation: <https://pig.apache.org/>
4. Reference project: <https://github.com/SiddhiPrabhu1995/YouTube-Data-Analysis-using-Hadoop>
5. Project GitHub Repository: <https://github.com/Sharif1023/BDAL-Final>